

Collection App ERP Integration

Technical Integration Scope and Requirements

Global IT Collection Application

Prepared for stakeholder review and approval

Document Purpose: This document is intended to align the next phase scope before development starts. It describes what will be built, why it matters, and how the different teams will use it.

1. Purpose

This document defines the ERP integration scope required for the next phase of the Collection App. It is intended for the ERP integration owner and the Collection App development team.

The integration must support an initial full data pull, real-time webhooks for ERP events, daily aging snapshots, document storage for invoice PDFs, and reconciliation between ERP and the Collection App.

2. Integration Principles

- ERP remains the source of truth for invoices, payments, customers, sales orders, official invoice PDFs, and financial balances.
- Collection App will store synchronized copies needed for analytics, operations, reminders, follow-ups, Welcome Call, and customer profiles.
- The system must support idempotent updates. Re-sending the same ERP event should not duplicate records.
- All records should carry ERP identifiers so they can be traced back to ERP.
- Large documents such as invoice PDFs should be stored in MinIO, while the database stores metadata and object references.

3. Recommended Integration Architecture

High-Level Architecture

ERP Full Sync API -> Collection Integration Layer -> Staging/Raw Tables -> Unified Tables -> Analytics and Operations
ERP Webhooks -> Collection Integration Layer -> Upsert Records -> Trigger Workflows -> Update Operational Views
Scheduled Aging Pull -> Snapshot Tables -> Date Comparison, DSO, Management Dashboards
Invoice PDF Source -> Collection App -> MinIO Storage -> Invoice Metadata Link

4. Integration Modes

Mode	Direction	Purpose	Timing
Initial Full Pull	Collection App pulls from ERP or ERP exposes one export endpoint.	Create the starting base for customers, invoices, payments, PDFs, products, contacts, sales orders, and aging.	Initial go-live and on-demand rebuild.
Real-Time Webhooks	ERP calls Collection App webhook endpoints.	Notify Collection App when a business event happens in ERP.	Immediately after ERP action.
Daily Scheduled Aging Pull	Collection App pulls ERP aging report.	Create official daily snapshot for analytics and comparison.	Daily, configurable time.
Reconciliation Pull	Collection App pulls summary or detailed data.	Detect missing invoices, missing payments, or total mismatches.	Daily or weekly.

5. Required ERP Data Domains

Domain	Full Pull	Webhook	Main Usage
Customers	Required	Required	Customer profile, Welcome Call, assignment, reporting.
Customer Contacts	Required	Required	Welcome Call, reminders, invoice delivery, escalation.
Sales Orders	Required for open/recent orders	Required	Add-on confirmation trigger and pre-invoice visibility.
Invoices	Required	Required	Invoice list, outstanding, reminders, product segmentation, DSO.
Invoice PDFs	Required for open invoices if available	Required	Official PDF storage in MinIO and customer reminder attachments.
Payments	Required	Required	Paid/partial/outstanding status, PTP closure, collection rate.
Aging Reports	Required	Scheduled pull preferred	Daily snapshots, comparison, DSO, dashboards.
Products / Items	Required	Optional/Required if ERP changes often	Product grouping and product-level analytics.
Credit / Debit Notes	Required if used by Finance	Required if used	Outstanding accuracy and invoice adjustments.
Write-offs / Adjustments	Required if used by Finance	Required if used	Financial accuracy and reconciliation.
KAM / Sales Ownership	Required	Required if updated in ERP	Sales and sales lead visibility and assignment context.
Service Activation / Delivery	Required if available	Required if available	Welcome Call timing, first invoice review, customer profile.

6. Full Pull / Baseline Sync Requirement

The ERP side should provide one full-base API, or a coordinated set of APIs, to allow the Collection App to create the starting base. The preferred approach is one orchestration endpoint that returns or triggers all required base domains.

Requirement	Details
Endpoint Option A	GET /erp/collection/base-sync?asOfDate=YYYY-MM-DD
Endpoint Option B	Separate ERP APIs for customers, contacts, invoices, payments, PDFs, aging, sales orders, and products.
Data Volume	Must support pagination, date filters, and retry.
Identifiers	Each object must include stable ERP IDs such as customer_code, invoice_no, payment_id, sales_order_no.
Re-run Support	The same full sync can be re-run without creating duplicates.
Audit	Collection App should store sync run ID, source, start time, end time, row counts, success/failure, and errors.

7. Webhook Event Envelope

All ERP webhooks should use a consistent wrapper. This makes retry, traceability, and idempotency easier.

Field	Required	Description
eventId	Yes	Unique ERP event ID. Used for idempotency.
eventType	Yes	Examples: customer.created, invoice.submitted, payment.created.
occurredAt	Yes	ERP event timestamp in ISO format.
sourceSystem	Yes	ERP system name/environment.
erpCompany	Optional	ERP company or business unit if available.
data	Yes	Event payload.
metadata	Optional	User, branch, correlation ID, retry count, or extra context.

Webhook Processing Rules

- Validate signature or token.
- Check eventId. If already processed, return success without duplicating data.
- Validate required fields.
- Upsert data using ERP identifiers.
- Trigger related workflow if applicable, such as Welcome Call, Add-on Confirmation, PTP check, or reminder update.
- Log processing result and errors.

8. Webhook Events Required

Event Type	When ERP Sends It	Collection App Action
customer.created	New customer is created in ERP.	Create customer profile and trigger New Customer Welcome Call.
customer.updated	Customer master data changes.	Update customer profile and contacts.
contact.created / contact.updated	Billing or customer contact changes.	Update contact details for reminders and calls.
sales_order.created_draft	Draft Sales Order created for existing customer add-on.	Trigger Existing Customer Add-on Confirmation.
sales_order.submitted	Sales Order submitted/approved.	Update sales order status and related add-on tracking.
sales_order.cancelled	Sales Order cancelled.	Close or cancel related add-on task if not invoiced.
invoice.created / invoice.submitted	Invoice is created/submitted in ERP.	Upsert invoice, update customer outstanding, prepare reminder data.
invoice.updated	Invoice data changes.	Update due date, amount, status, product lines, or balance.
invoice.cancelled	Invoice cancelled.	Mark invoice as cancelled and update balances.
invoice.pdf_ready	Official invoice PDF is generated or updated.	Download/store PDF in MinIO and link it to invoice.
payment.created / payment.submitted	Payment is posted in ERP.	Update paid amount, outstanding, PTP status, and collection rate.
payment.cancelled / reversed	Payment is cancelled or reversed.	Reverse payment effect and recalculate outstanding.
credit_note.created / allocated	Credit note is created or allocated.	Update invoice balance and financial adjustments.
debit_note.created	Debit note is created.	Update customer balance and invoice-related view.
item.created / item.updated	Product/item data changes.	Update product mapping source and product grouping suggestions.
kam_or_owner.updated	Sales/KAM ownership changes in ERP.	Update sales ownership context.

9. Data Objects and Minimum Fields

9.1 Customer

Field	Required	Notes
customerCode	Yes	Stable ERP customer identifier.
customerName	Yes	Customer display name.
status	Yes	Active, inactive, suspended, etc.
accountType	Optional	Global, Afaqy, or other.
kamName / kamCode	Recommended	Commercial owner if available.
salesLead	Optional	If ERP stores hierarchy.
billingEmail	Recommended	Default reminder recipient.
phone	Recommended	Welcome Call and follow-up.
createdAt	Yes	Used for Welcome Call trigger.

9.2 Sales Order

Field	Required	Notes
salesOrderNo	Yes	Stable ERP Sales Order ID.
customerCode	Yes	Links order to customer.
status	Yes	Draft, submitted, cancelled.
isAddon	Recommended	If available; otherwise Collection App infers existing customer.
orderDate	Yes	Creation date.
items	Yes	Product/service lines.
totalAmount	Recommended	Expected billing value.
createdBy	Recommended	Sales Support user if available.

9.3 Invoice

Field	Required	Notes
invoiceNo	Yes	Stable ERP invoice number.
customerCode	Yes	Links invoice to customer.
invoiceDate	Yes	Used for aging and DSO.
dueDate	Recommended	Used for overdue calculations.
status	Yes	Draft, submitted, paid, partial, cancelled, overdue.
totalAmount	Yes	Invoice gross/total amount.
paidAmount	Recommended	If ERP provides.
outstandingAmount	Yes	Current unpaid amount.
currency	Yes	Expected SAR but should be explicit.
salesOrderNo	Optional	Important for add-on traceability.
lineItems	Yes	Needed for product segmentation.

9.4 Invoice PDF

Field	Required	Notes
invoiceNo	Yes	Link to invoice.
pdfUrl or fileReference	Yes	Preferred: downloadable URL or document ID. Avoid sending huge binary in webhook.
pdfGeneratedAt	Recommended	Timestamp.
zatcaReference	Optional	If available from ERP.
checksum	Recommended	For file integrity.
version	Optional	If PDFs can be regenerated.

9.5 Payment

Field	Required	Notes
paymentId	Yes	Stable ERP payment entry ID.
paymentDate	Yes	Date of payment.
customerCode	Yes	Customer.
amount	Yes	Payment amount.
currency	Yes	Currency.
allocatedInvoices	Yes	Invoice allocations with amount per invoice.
status	Yes	Submitted, cancelled, reversed.
referenceNo	Optional	Bank/reference number.

9.6 Aging Snapshot

Field	Required	Notes
snapshotDate	Yes	Official aging date.
customerCode	Yes	Customer.
accountType	Optional	Global, Afaqy, etc.
totalOutstanding	Yes	Total outstanding.
aging0_30	Yes	Bucket amount.
aging31_60	Yes	Bucket amount.
aging61_90	Yes	Bucket amount.
aging91_120	Yes	Bucket amount.
aging121_200	Recommended	Bucket amount if ERP supports.
aging201_300	Recommended	Bucket amount if ERP supports.
aging301_400	Recommended	Bucket amount if ERP supports.
aging401_500	Recommended	Bucket amount if ERP supports.
aging501Above	Recommended	Bucket amount if ERP supports.

10. Product Segmentation Requirement

ERP invoice line items must include product/service names. Collection App will group detailed ERP product names into business categories using configurable rules.

ERP Product Name	Grouping Rule Example	Final Category
DIA 100 GB	Contains DIA	DIA
DIA 500 GB	Contains DIA	DIA
M2M SIM Package	Contains M2M or SIM mapping	M2M
Smart VPN Service	Contains Smart VPN	Smart VPN

11. Welcome Call and Add-on Confirmation Triggers

ERP Event	Condition	Workflow Created
customer.created	Customer is new in ERP.	New Customer Welcome Call.
sales_order.created_draft	Existing customer add-on, Sales Order created as draft.	Existing Customer Add-on Confirmation.
invoice.created without sales_order_no	Add-on invoiced directly without Sales Order.	No automatic pre-invoice Welcome Call trigger. This is an agreed exception.

Exception Rule: The direct invoice exception should be logged or visible if detected, but it should not create the pre-invoice add-on confirmation workflow because the invoice already exists and there was no Sales Order event to capture.

12. Invoice PDF Storage in MinIO

PDF Handling Flow

ERP generates official invoice PDF.
ERP sends invoice.pdf_ready webhook with PDF URL/document reference.
Collection App downloads the PDF or requests it from ERP.
Collection App stores the file in MinIO.
Database stores invoice number, ERP document reference, MinIO object key, checksum, version, and timestamps.
Reminder module uses the stored PDF when sending customer statements.

13. Update Strategy

Data / Event	Update Strategy	Reason
New customer	Webhook immediate	Triggers Welcome Call and customer profile creation.
Sales order	Webhook immediate	Triggers add-on confirmation before invoicing.
Invoice	Webhook immediate	Needed for operational invoice visibility and reminders.
Invoice PDF	Webhook immediate	Needed for official customer communication.
Payment	Webhook immediate	Updates balances and may close PTP.
Contact updates	Webhook immediate	Keeps reminder and call details accurate.
Aging report	Daily scheduled pull	Creates stable official snapshot for analytics.
DSO and comparison	Based on daily snapshots plus invoice/payment data	Needs consistent reporting periods.
Reconciliation	Daily or weekly scheduled pull	Detects missed events and mismatches.

14. Security and Reliability

Area	Requirement
Authentication	Webhook endpoints must use token, signature, mTLS, or agreed secure authentication method.
Idempotency	eventId must be unique and stored. Repeated eventId returns success without duplicate processing.
Retry	ERP should retry failed webhooks with backoff. Collection App should return clear success/failure codes.
Logging	Every sync and webhook should be logged with request ID, status, received time, processed time, and error details.
Validation	Invalid payloads should be rejected and logged with clear missing/invalid fields.
Ordering	If events arrive out of order, Collection App should use ERP timestamps and current ERP status to avoid incorrect regression.
Reconciliation	Scheduled checks should detect missing or inconsistent records.

15. Suggested Collection App Endpoint Names

Endpoint	Purpose
POST /integration/erp/webhooks	Generic webhook endpoint using eventType in the event envelope.
POST /integration/erp/full-sync/start	Start or request baseline sync from ERP.
GET /integration/erp/full-sync/status/:runId	Check sync status.
POST /integration/erp/aging/snapshot	Receive or trigger aging snapshot import.
POST /integration/erp/reconciliation/start	Start reconciliation job.
GET /integration/erp/reconciliation/status/:runId	Check reconciliation results.

Implementation Option: A generic webhook endpoint is easier to maintain if the ERP integration can send a consistent event envelope. Dedicated endpoints can also be used if preferred by the ERP team.

16. Integration Testing Scenarios

- Full sync imports customers, contacts, invoices, payments, PDFs, products, sales orders, and aging without duplicates.
- New ERP customer triggers a New Customer Welcome Call task.
- Draft Sales Order for existing customer triggers Existing Customer Add-on Confirmation task.
- Invoice created webhook creates or updates invoice and product line items.
- PDF ready webhook stores PDF in MinIO and links it to invoice.
- Payment created webhook updates outstanding and closes fulfilled PTP if applicable.
- Payment reversal correctly restores outstanding balance.
- Daily aging snapshot creates a new snapshot date for comparison.
- Repeated webhook with same eventId does not duplicate data.
- Reconciliation detects missing invoice or payment records.

17. Open Technical Decisions

Decision	Owner / Needed From ERP Side
Full pull method	Confirm whether ERP will expose one base export API or separate APIs.
Webhook authentication	Confirm token/signature/mTLS approach.
PDF delivery method	Confirm whether ERP sends PDF URL, document ID, or binary. Preferred: URL/document ID.
Aging report format	Confirm available aging buckets and whether report is customer-level or invoice-level.
Product fields	Confirm invoice line fields available for product grouping.
Sales Order event availability	Confirm draft Sales Order webhook can be sent when created.
Direct invoice add-ons	Confirm these cannot be captured pre-invoice and will be excluded from automatic Welcome Call trigger.
Historical range	Confirm how many months/years of invoices and payments should be pulled initially.
Environment	Confirm development, UAT, and production ERP endpoints.

18. Final Summary

The required ERP integration is a hybrid model: one baseline full pull to copy the starting base, real-time webhooks for ERP actions, daily aging snapshots for reporting, and reconciliation to protect accuracy. This allows the Collection App to support immediate operational work while keeping management analytics stable and auditable.